

# Primer on random processes

## MG3416–Advanced Structural Acoustics – Lecture #1

### part A

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# Probability space

## Formal definition

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- We model a real-world process, or "experiment", or "system", consisting of **states** occurring randomly.
- It is mathematically represented by a triplet  $(\Omega, \mathcal{F}, P)$  of which elements are ultimately specified by the modeler:
  - $\Omega$  is the set of outcomes, or **sample space**, or **universe**, constituted by the states of the causes which influence the states of the system;
  - $\mathcal{F}$  is a  **$\sigma$ -algebra**, *i.e.* a set of all the events the modeler considers. An **event** is a set of zero or more outcomes, thus it is a subset of the sample space;
  - $P$  is a function returning an event's probability, or a **probability measure**.

# Probability space

## $\sigma$ -algebra

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## Definition

A collection  $\mathcal{F}$  of subsets of  $\Omega$  is a  $\sigma$ -algebra, or  $\sigma$ -field, if:

- 1  $\emptyset \in \mathcal{F}$  and  $\Omega \in \mathcal{F}$ ;
- 2  $a \in \mathcal{F} \Rightarrow a^c = \Omega \setminus a \in \mathcal{F}$ ;
- 3 If  $\{a_n\}_{n=1,2,\dots}$  is an arbitrary countable family of events in  $\mathcal{F}$ , then  $\bigcup_{n \in \mathbb{N}} a_n \in \mathcal{F}$ .

- **Example #1:** a fair coin,  $\Omega = \{1, 2\}$ . Then the  $\sigma$ -algebra  $\mathcal{F} = 2^\Omega = \{\emptyset, \{1\}, \{2\}, \Omega\}$  contains  $2^2 = 4$  events.
- **Example #2:** a fair die,  $\Omega = \{1, 2, 3, 4, 5, 6\}$ . Then the  $\sigma$ -algebra  $\mathcal{F} = 2^\Omega = \{\emptyset, \{1\}, \{2\}, \dots, \{6\}, \{1, 2\}, \{3, 4\}, \{5, 6\}, \{1, 3\}, \dots, \{1, 3, 5\}, \{2, 4, 6\}, \{1, 3, 4, 5\}, \dots, \Omega\}$  contains  $2^6 = 64$  events.
- **Example #3:** a fair die,  $\Omega = \{1, 2, 3, 4, 5, 6\}$ . The sets  $\mathcal{F} = \{\emptyset, \{2, 4, 6\}, \{1, 3, 5\}, \Omega\}$ ,  $\mathcal{F} = \{\emptyset, \{2, 5\}, \{1, 3, 4, 6\}, \Omega\}$ ... are also suitable  $\sigma$ -algebras.

# Probability space

## Probability measure

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### Definition

*A probability measure is a positive, bounded measure the total mass of which is 1:  $P(\Omega) = 1$ .*

- 1** *A positive measure  $m$  is a function  $m : \Omega \rightarrow [0, +\infty]$  s.t. for countably many disjoint events  $\{a_n\}_{n=1,2,\dots} \in \mathcal{F}$ ,  $\bigcap_n a_n = \emptyset$ ,  $\bigcup_n a_n \in \mathcal{F}$ , then*

$$m\left(\bigcup_n a_n\right) = \sum_n m(a_n);$$

- 2** *It is bounded if  $m(\Omega) < +\infty$ ;*
- 3** *It is a probability measure if  $m(\Omega) = 1$ .*

# Random variable

## Definitions

### Definition

*A random variable is a measurable function  $X : \Omega \rightarrow \Omega'$  which maps a combination of outcomes from a sample space  $\Omega$  to a combination of consequences in a state space  $\Omega'$ .*

### Definition

*A function  $f : \Omega \rightarrow \Omega'$  is measurable if  $\forall a' \in \Omega', f^{-1}(a')$  is measurable in  $\Omega$ :  $\forall a' \in \mathcal{F}', f^{-1}(a') = \{a \in \Omega; f(a) \in a'\} \in \mathcal{F}$ , where  $\mathcal{F}, \mathcal{F}'$  are  $\sigma$ -algebras defined on  $\Omega, \Omega'$  respectively.*

- **Example #1:** let  $\Omega$  be the sample space describing the real-world system "tossing a fair coin":  $\Omega = \{"temperature", "pressure", "initial position", "initial speed", etc.\}$ . Then  $\Omega' = \{1, 2\}$ .
- **Example #2:** let  $\Omega$  be the sample space describing a real-world system "roll of dice":  $\Omega = \{"temperature", "pressure", "initial position", "initial speed", etc.\}$ . Then  $\Omega' = \{1, 2, 3, 4, 5, 6\}$ .

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# Random variable

## Causality principle

- Let  $X : \Omega \rightarrow \Omega'$  be a random variable, then  $\Omega'$  is endowed with the  $\sigma$ -algebra  $\mathcal{F}'$  and the probability measure  $P'$  inherited from  $\mathcal{F}$  and  $P$  by the **causality principle**:

$$\forall a' \in \mathcal{F}', \quad P'(a') = P(X^{-1}(a')).$$

$P'$  is now the probability measure of  $X$  on  $\Omega'$ .

- One can always find states  $a$  in  $\mathcal{F}$  such that  $X(a)$  is in  $\mathcal{F}'$ ! (that's measurability).
- The original sample space  $\Omega$  (the system and its environment in the foregoing examples) is often suppressed, since it is mathematically hard to describe.
- **Examples:** The possible values of the random variable "coin toss" or "dice roll" are then treated as a sample space  $\Omega'$  constituted by numerical quantities—which is easier to deal with.



# Stochastic process

## Definition and terminology

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## Definition

*A stochastic process:*

$$\begin{aligned}\mathbf{X} : T &\rightarrow \mathbb{R}^n \\ t &\mapsto \mathbf{X}_t\end{aligned}$$

*is a function from  $T$  into the set of random variables defined on  $(\Omega, \mathcal{F}, P)$  with values in  $\mathbb{R}^n \equiv \Omega'$ .  $(\mathbf{X}_t, t \in T)$  is a collection of random variables with values in  $\mathbb{R}^n$ . Therefore:*

- *If  $t \in T$  is frozen:  $\mathbf{X}_t$  is a random variable defined on  $(\Omega, \mathcal{F}, P)$  with values in  $\mathbb{R}^n$ ;*
- *If  $a \in \Omega$  is frozen:  $t \mapsto \mathbf{X}_t(a) = \mathbf{x}(t)$  is a trajectory, or a **sample path**;*
- *If both  $t \in T$  and  $a \in \Omega$  are frozen:  $a, t \mapsto \mathbf{X}_t(a)$  is an outcome, or a **realization**, of the random variable  $\mathbf{X}_t$ .*

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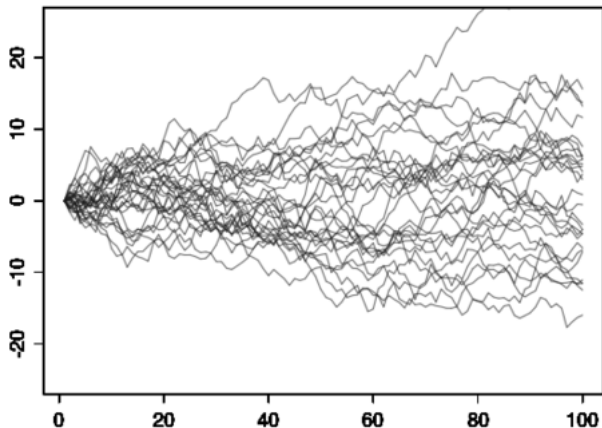
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Stochastic process indexed on  $\mathbb{R}_+$ .

# Stochastic process

## Remarks

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- $(\mathbf{X}_t, t \in T)$  is called "the stochastic process defined on  $(\Omega, \mathcal{F}, P)$ , indexed on  $T$ , with values in  $\mathbb{R}^n$ ".
- The indexation set  $T$  could be any set, countable or not, bounded or not, *etc.*
- However we will only consider the real line  $T = \mathbb{R}$  in the subsequent lectures, since  $t$  will be a time.
- **Examples:** earthquake, wind, turbulence, swell, ocean surge... can be modeled by stochastic processes, where  $\Omega = \{\text{"the states of the environment"}\}$  is very difficult to figure out, but  $\Omega' = \{\text{"vectors of } \mathbb{R}^3\}\}$  is much easier to handle.

# Stochastic process

## Probability law

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- For  $t \in \mathbb{R}$  fixed,  $\mathbf{X}_t$  is a  $\mathbb{R}^n$ -valued random variable defined on  $(\Omega, \mathcal{F}, P)$ , of which **probability law**  $P_{\mathbf{X}}$  is parameterized by  $t$  and defined by:

$$\begin{aligned} P_{\mathbf{X}}(B; t) &= P'(\mathbf{X}_t \in B) \\ &= P(\mathbf{X}_t^{-1}(B) \in \mathcal{F}), \quad \forall B \in \mathcal{F}'. \end{aligned}$$

- The most common choice for the  $\sigma$ -algebra  $\mathcal{F}'$  is the Borel  $\sigma$ -algebra  $\mathcal{B}(\mathbb{R}^n)$ —the one generated by the collection of all open sets in  $\mathbb{R}^n$ .
- The **distribution function**  $F_{\mathbf{X}}(\mathbf{x}; t) : \mathbb{R}^n \rightarrow [0, 1]$ :

$$\begin{aligned} P_{\mathbf{X}}(d\mathbf{x}; t) &= dF_{\mathbf{X}}(\mathbf{x}; t), \\ F_{\mathbf{X}}(\mathbf{x}; t) &= P'(\mathbf{X}_t \in ]-\infty, x_1] \times \cdots \times ]-\infty, x_n]). \end{aligned}$$

# Stochastic process

## Joint and marginal probability laws

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- **Joint probability laws:** The two-fold probability measure  $P_{\mathbf{X}\mathbf{Y}}(d\mathbf{x}d\mathbf{y}; t, t') = dF_{\mathbf{X}\mathbf{Y}}(\mathbf{x}, \mathbf{y}; t, t')$  with:

$$F_{\mathbf{X}\mathbf{Y}}(\mathbf{x}, \mathbf{y}; t, t') = P'(\mathbf{X}_t \in ]-\infty, x_1] \times \cdots \times ]-\infty, x_n] \\ \text{and } \mathbf{Y}_{t'} \in ]-\infty, y_1] \times \cdots \times ]-\infty, y_n]).$$

- **Marginal probability laws:** Let  $I$  be the set of all finite, unordered parts of  $\mathbb{R}$  of the form  $I \ni i = \{t_1, t_2, \dots, t_k\}$ . Let  $\mathbf{X}^{(i)} = (\mathbf{X}_{t_1}, \mathbf{X}_{t_2}, \dots, \mathbf{X}_{t_k})$  be the  $\mathbb{R}^{n \times k}$ -valued random variable constructed from the part  $i$ , then the collection of marginal probability laws of the process  $(\mathbf{X}_t, t \in \mathbb{R})$  is  $\{P_{\mathbf{X}^{(i)}}(d\mathbf{x}_1 d\mathbf{x}_2 \cdots d\mathbf{x}_k), i \in I\}$ .

# Stochastic process

## Second-order stochastic process

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### Definition

$(\mathbf{X}_t, t \in \mathbb{R})$  is a second-order stochastic process if:

$$\mathbb{E}\{\|\mathbf{X}_t\|^2\} < +\infty, \quad \forall t \in \mathbb{R},$$

where  $\|\mathbf{x}\| = \sqrt{\sum_{j=1}^n x_j^2}$  and  $\mathbb{E}\{f(\mathbf{X}_t)\} = \int_{\mathbb{R}^n} f(\mathbf{x}) P_{\mathbf{X}}(d\mathbf{x}; t)$ .

- The set  $L^2(\Omega, \mathbb{R}^n)$  of  $\mathbb{R}^n$ -valued second-order random variables defined on  $(\Omega, \mathcal{F}, P)$  is an Hilbert space, with the scalar product  $((\mathbf{X}, \mathbf{Y}))_2 = \mathbb{E}\{(\mathbf{X}, \mathbf{Y})\}$  and the norm  $\|\mathbf{X}\| = \sqrt{((\mathbf{X}, \mathbf{X}))_2} = \sqrt{\mathbb{E}\{\|\mathbf{X}\|^2\}}$ .
- **Interests:** the same as for  $L^2(\mathbb{R}^n)$ ! (separability, uniform convexity).

# Stochastic process

## Mean and auto-correlation functions

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### Definition

Let  $(\mathbf{X}_t, t \in \mathbb{R})$  be a  $\mathbb{R}^n$ -valued, second-order stochastic process defined on  $(\Omega, \mathcal{F}, P)$  and indexed on  $\mathbb{R}$ .

- Its mean function  $\boldsymbol{\mu}_{\mathbf{X}} : \mathbb{R} \rightarrow \mathbb{R}^n$  is:

$$\boldsymbol{\mu}_{\mathbf{X}}(t) = \int_{\mathbb{R}^n} \mathbf{x} P_{\mathbf{X}}(d\mathbf{x}; t) = \mathbb{E}\{\mathbf{X}_t\}.$$

The process is centered if  $\boldsymbol{\mu}_{\mathbf{X}}(t) = \mathbf{0}, \forall t \in \mathbb{R}$ .

- Its auto-correlation function  $\mathbf{R}_{\mathbf{X}} : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}^{n \times n}$  is:

$$\mathbf{R}_{\mathbf{X}}(t, t') = \iint_{\mathbb{R}^n \times \mathbb{R}^n} \mathbf{x} \otimes \mathbf{y} P_{\mathbf{X}\mathbf{X}}(d\mathbf{x} d\mathbf{y}; t, t') = \mathbb{E}\{\mathbf{X}_t \otimes \mathbf{X}_{t'}\}.$$

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# Mean-square stationary stochastic process

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### Definition

Let  $(\mathbf{X}_t, t \in \mathbb{R})$  be a  $\mathbb{R}^n$ -valued stochastic process defined on  $(\Omega, \mathcal{F}, P)$  and indexed on  $\mathbb{R}$ . It is:

- stationary if  $\forall u \in \mathbb{R}$ ,  $(\mathbf{X}_{t+u}, t \in \mathbb{R})$  and  $(\mathbf{X}_t, t \in \mathbb{R})$  have the same marginal probability laws;
  - mean-square stationary if it is second-order,  $\boldsymbol{\mu}_{\mathbf{X}}(t) = \boldsymbol{\mu}_{\mathbf{X}}$  (independent of time), and  $\mathbf{R}_{\mathbf{X}}(t, t') = \mathbf{R}_{\mathbf{X}}(t - t')$ .
- 
- If  $(\mathbf{X}_t, t \in \mathbb{R})$  is stationary then  $P_{\mathbf{X}}(d\mathbf{x}; t) = P_{\mathbf{X}}(d\mathbf{x})$  (independent of time) and  $P_{\mathbf{X}\mathbf{X}}(d\mathbf{x}d\mathbf{y}; t, t') = P_{\mathbf{X}\mathbf{X}}(d\mathbf{x}d\mathbf{y}; t - t')$ .
  - Stationary  $\Rightarrow$  mean-square stationary, but the reverse is not true.

# Mean-square stationary stochastic process

## Stationarity

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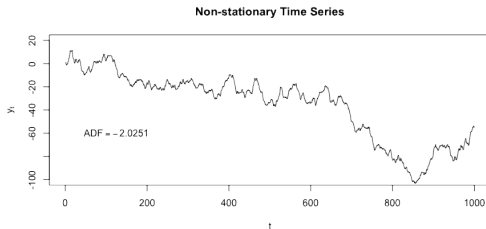
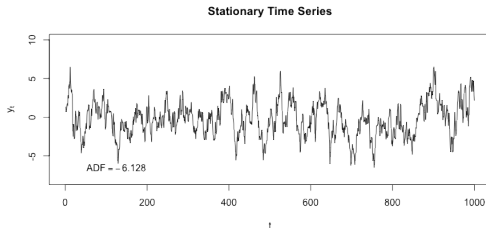
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Stationary *v.s.* non-stationary processes.

# Mean-square stationary stochastic process

## Spectral measure

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- **Data:**  $(\mathbf{X}_t, t \in \mathbb{R})$  a  $\mathbb{R}^n$ -valued, second-order stochastic process defined on  $(\Omega, \mathcal{F}, P)$ , indexed on  $\mathbb{R}$ , and mean-square stationary.
- **Hypothesis:**  $t \mapsto \mathbf{R}_X(t)$  is continuous on  $\mathbb{R}$ .

### Theorem (Bochner)

*Then there exists an Hermitian, positive, and bounded measure  $\mathbf{M}_X \in \mathbb{C}^{n \times n}$ , called the spectral measure of the  $\mathbb{R}^n$ -valued stochastic process  $(\mathbf{X}_t, t \in \mathbb{R})$ , s.t.:*

$$\mathbf{R}_X(t) = \int_{\mathbb{R}} e^{i\omega t} \mathbf{M}_X(d\omega).$$

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## Power spectral density

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### Definition

*The spectral density  $\mathbf{S}_{\mathbf{X}}$  is the density of the spectral measure  $\mathbf{M}_{\mathbf{X}}$  if it exists:*

$$\mathbf{M}_{\mathbf{X}}(d\omega) = \mathbf{S}_{\mathbf{X}}(\omega)d\omega.$$

- It is Hermitian, positive, and integrable (since  $\mathbf{M}_{\mathbf{X}}(\mathbb{R}) < +\infty$ ). Then  $\lim_{|t| \rightarrow +\infty} \mathbf{R}_{\mathbf{X}}(t) = 0$  (as the Fourier transform of an integrable function).
- A necessary condition:  $\boldsymbol{\mu}_{\mathbf{X}} = \mathbf{0}$ .
- A sufficient condition:  $\lim_{|t| \rightarrow +\infty} \mathbf{R}_{\mathbf{X}}(t) = 0$ .
- Let  $B \subseteq \mathbb{R}$ ,  $B \mapsto \text{Tr } \mathbf{M}_{\mathbf{X}}(B)$  is the **power spectral measure** on  $B$ , and if  $\mathbf{S}_{\mathbf{X}}$  exists,  $\omega \mapsto \text{Tr } \mathbf{S}_{\mathbf{X}}(\omega)$  is the **power spectral density** (PSD) of the process.

# Mean-square stationary stochastic process

## Mean-square derivative

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- **Data:**  $(\mathbf{X}_t, t \in \mathbb{R})$  a  $\mathbb{R}^n$ -valued, second-order stochastic process defined on  $(\Omega, \mathcal{F}, P)$ , indexed on  $\mathbb{R}$ .

### Definition

*Let  $t \in \mathbb{R}$ ,  $(\mathbf{X}_t, t \in \mathbb{R})$  is mean-square derivable at  $t$  iff the sequence of random variables  $\mathbf{X}_h = \frac{1}{h}(\mathbf{X}_{t+h} - \mathbf{X}_t)$  converges in  $L^2(\Omega, \mathbb{R}^n)$  as  $h \rightarrow 0$ . That limit is denoted by  $\dot{\mathbf{X}}_t$  s.t.:*

$$\lim_{h \rightarrow 0} \|\mathbf{X}_h - \dot{\mathbf{X}}_t\| = 0.$$

# Mean-square stationary stochastic process

## Second-order properties of the derivative

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- A sufficient condition:  $\frac{\partial^2 \mathbf{R}_X}{\partial t \partial t'}$  exists for all  $(t, t')$  in  $\mathbb{R} \times \mathbb{R}$ .
- Then  $(\dot{\mathbf{X}}_t, t \in \mathbb{R})$  is a  $\mathbb{R}^n$ -valued, second-order stochastic process defined on  $(\Omega, \mathcal{F}, P)$ , indexed on  $\mathbb{R}$ .  
In addition:

$$\mu_{\dot{\mathbf{X}}}(t) = \frac{d\mu_X}{dt}(t),$$

$$\mathbf{R}_{\dot{\mathbf{X}}}(t, t') = \frac{\partial^2 \mathbf{R}_X}{\partial t \partial t'}(t, t'),$$

$$\mathbf{R}_{\mathbf{X}\dot{\mathbf{X}}}(t, t') = \mathbb{E}\{\mathbf{X}_t \otimes \dot{\mathbf{X}}_{t'}\} = \frac{\partial \mathbf{R}_X}{\partial t'}(t, t'),$$

$$\mathbf{R}_{\dot{\mathbf{X}}\mathbf{X}}(t, t') = \mathbb{E}\{\dot{\mathbf{X}}_t \otimes \mathbf{X}_{t'}\} = \frac{\partial \mathbf{R}_X}{\partial t}(t, t').$$

# Mean-square stationary stochastic process

## Derivative of a mean-square stationary process

- A necessary and sufficient condition:

$$\int_{\mathbb{R}} \omega^2 \operatorname{tr} \mathbf{M}_{\mathbf{X}}(d\omega) < +\infty.$$

- Then  $(\dot{\mathbf{X}}_t, t \in \mathbb{R})$  is a  $\mathbb{R}^n$ -valued, second-order stochastic process defined on  $(\Omega, \mathcal{F}, P)$ , indexed on  $\mathbb{R}$ , and mean-square stationary. In addition:

$$\mu_{\dot{\mathbf{X}}}(t) = \mathbf{0},$$

$$\mathbf{R}_{\dot{\mathbf{X}}}(t) = -\frac{d^2 \mathbf{R}_{\mathbf{X}}}{dt^2}(t),$$

$$\mathbf{R}_{\dot{\mathbf{X}}\mathbf{X}}(t) = \frac{d\mathbf{R}_{\mathbf{X}}}{dt}(t) = -\mathbf{R}_{\mathbf{X}\dot{\mathbf{X}}}(t),$$

$$\mathbf{M}_{\dot{\mathbf{X}}}(d\omega) = \omega^2 \mathbf{M}_{\mathbf{X}}(d\omega).$$

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# Filtering of a stochastic process

## Linear filter

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Probability  
Space

Probability  
triplet

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Stochastic  
process

Mean-  
square  
stationary  
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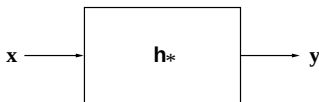
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- A linear convolution filter:

$$\mathbf{y}(t) = \int_{\mathbb{R}} \mathbf{h}(t - \tau) \mathbf{x}(\tau) d\tau$$

where:

- $t \mapsto \mathbf{h}(t)$  is the **impulse response function** in  $\mathbb{R}^{n \times m}$ ;
- $\omega \mapsto \widehat{\mathbf{h}}(\omega) = \int_{\mathbb{R}} e^{-i\omega t} \mathbf{h}(t) dt$  is the **frequency response function**;
- $p \mapsto \mathbb{H}(p) = \int_{-\infty}^{+\infty} e^{-pt} \mathbf{h}(t) dt$ ,  $p \in D_{\mathbb{H}} \subset \mathbb{C}$ , is the **transfer function**.

# Filtering of a stochastic process

## Stationary response

- **Data:**  $(\mathbf{X}_t, t \in \mathbb{R})$  a  $\mathbb{R}^m$ -valued, second-order stochastic process defined on  $(\Omega, \mathcal{F}, P)$ , indexed on  $\mathbb{R}$ , and mean-square stationary.
- If  $t \mapsto \mathbf{h}(t)$  is integrable or square integrable, then  $(\mathbf{Y}_t, t \in \mathbb{R})$  is a  $\mathbb{R}^n$ -valued, second-order stochastic process defined on  $(\Omega, \mathcal{F}, P)$ , indexed on  $\mathbb{R}$ , and mean-square stationary s.t.:

$$\boldsymbol{\mu}_{\mathbf{Y}} = \hat{\mathbf{h}}(0) \boldsymbol{\mu}_{\mathbf{X}},$$

$$\mathbf{R}_{\mathbf{Y}}(t) = \iint_{\mathbb{R} \times \mathbb{R}} \mathbf{h}(\tau) \mathbf{R}_{\mathbf{X}}(t + \tau' - \tau) \mathbf{h}(\tau')^{\top} d\tau d\tau',$$

$$\mathbf{M}_{\mathbf{Y}}(d\omega) = \hat{\mathbf{h}}(\omega) \mathbf{M}_{\mathbf{X}}(d\omega) \hat{\mathbf{h}}^*(\omega).$$

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